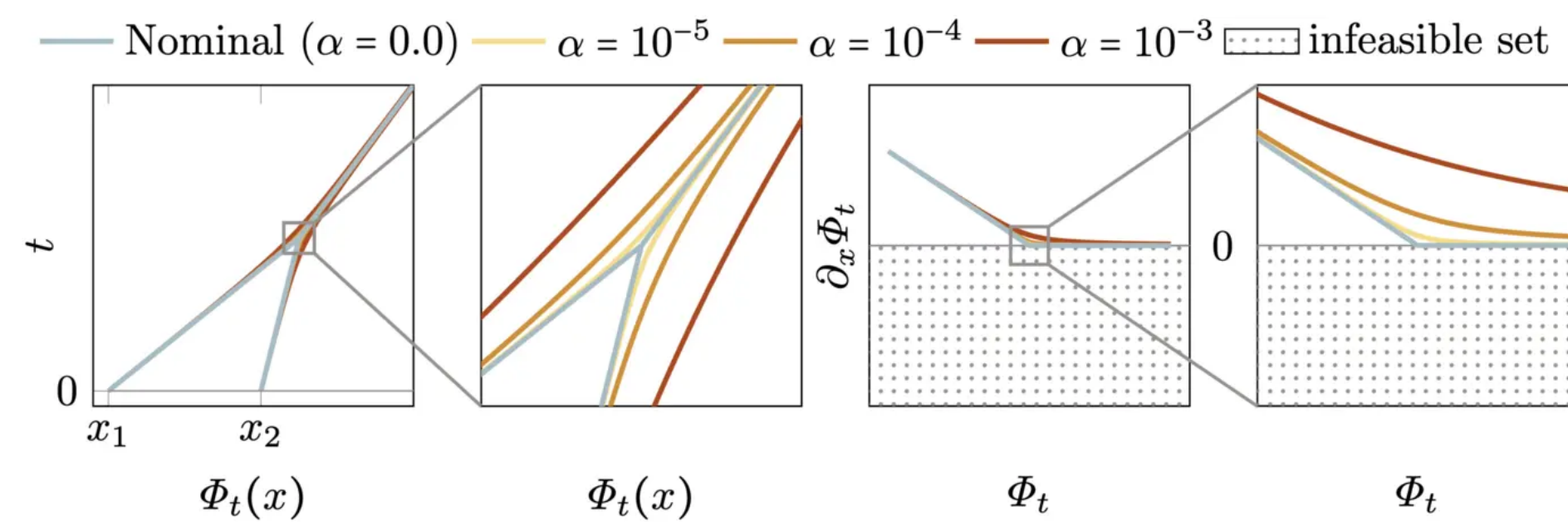


Compressible Euler & Shock Formation

- Compressible Euler flows develop shocks from smooth initial data.
- Shock formation corresponds to particle trajectories intersecting.
- Standard stabilization methods act after strong gradients form on the grid scale.



Information Geometric Regularization (IGR)

- Cao and Schäfer [1] introduce Information Geometric Regularization (IGR), a PDE-level regularization for compressible flow.
- IGR introduces an entropic pressure Σ that acts as a geometric barrier against particle trajectories intersecting.
- The pressure in the flux is augmented from $p \mapsto p + \Sigma$.

$$\partial_t \begin{pmatrix} \rho \\ \rho u \\ E \end{pmatrix} + \nabla \cdot \begin{pmatrix} \rho u \\ \rho u \otimes u + (p + \Sigma)I \\ (E + p + \Sigma)u \end{pmatrix} = 0.$$

- Σ is determined by an elliptic PDE constraint that depends on the instantaneous flow state.

$$\rho^{-1}\Sigma - \alpha \nabla \cdot (\rho^{-1} \nabla \Sigma) = \alpha (\text{tr}(Du)^2 + \text{tr}((Du)^2)).$$

IGR-DG Method

- Conserved variables and entropic pressure discretized with DG [2, 3].

$$U_h = \sum_j U_j(t) \phi_j(x), \quad \Sigma_h = \sum_j \gamma_j(t) \phi_j(x).$$

- Hyperbolic transport retains the conservative formulation.
- The DG discretized systems are

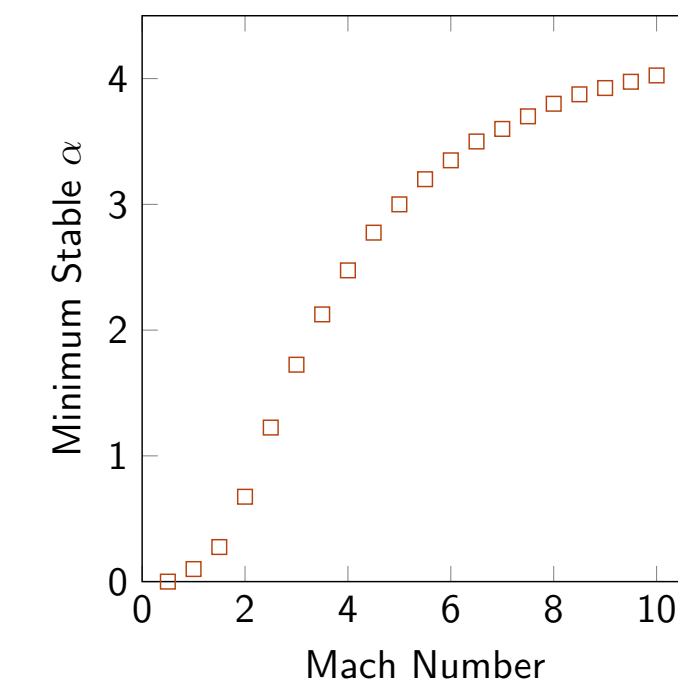
$$M \dot{U} = K(U) + B(U), \quad (M^\Sigma + K^\Sigma + B^\Sigma) \gamma = g^\Sigma.$$

- A sparse SPD elliptic system is solved via conjugate gradient.
- Sparse elliptic solve adds modest computational overhead.

IGR Regularization Strength

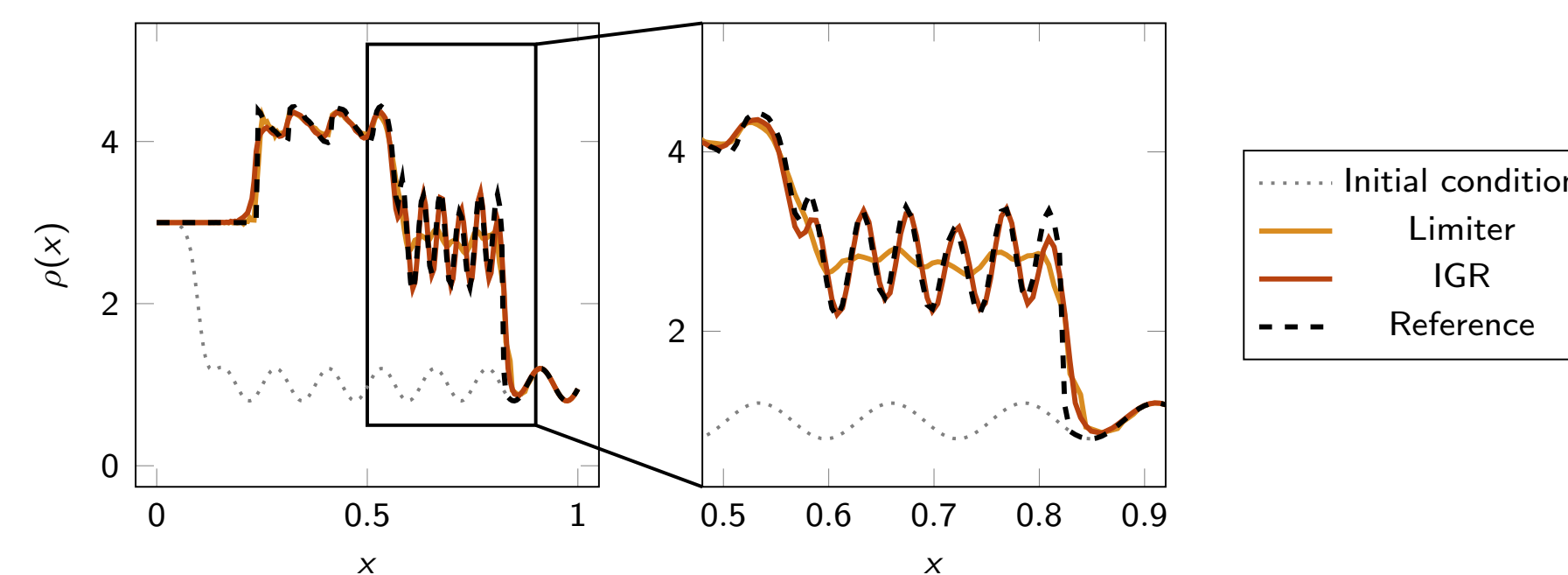
- α controls the regularization strength.
- Regularization strength scales as $\sqrt{\alpha} \sim \Delta x$.
- A minimum α value is required to stabilize shocks of a given strength, but larger α is typically selected to suppress oscillations.
- The minimum stable α increases with Mach number.
- Effective regularization is inversely proportional to polynomial order

$$\alpha_{\text{eff}} = \frac{4\alpha}{(p+1)^2}$$

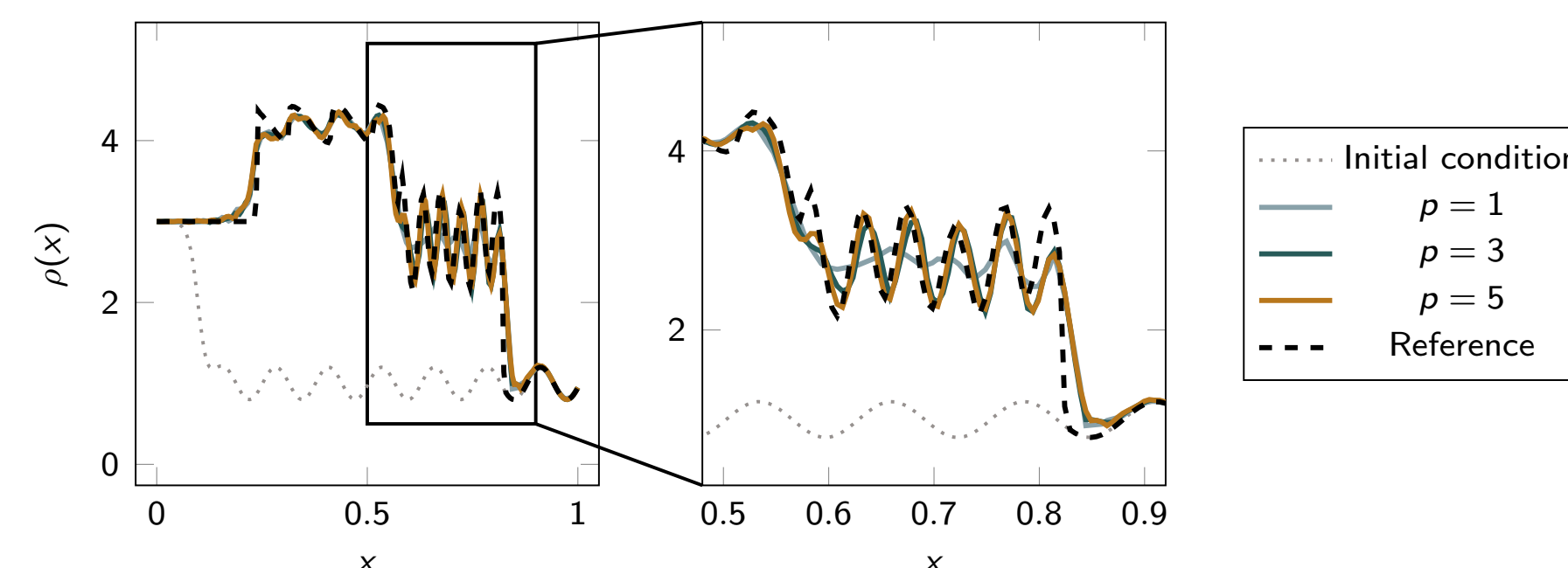


1D Numerical Results: Smoothed Shu–Osher

- The smoothed Shu–Osher problem tests shock interaction with oscillatory structure [4].
- IGR stabilizes shocks while preserving physically meaningful fine-scale structure in smooth regions.
- Limiter stabilization suppresses higher-order modes and damps post-shock oscillations.

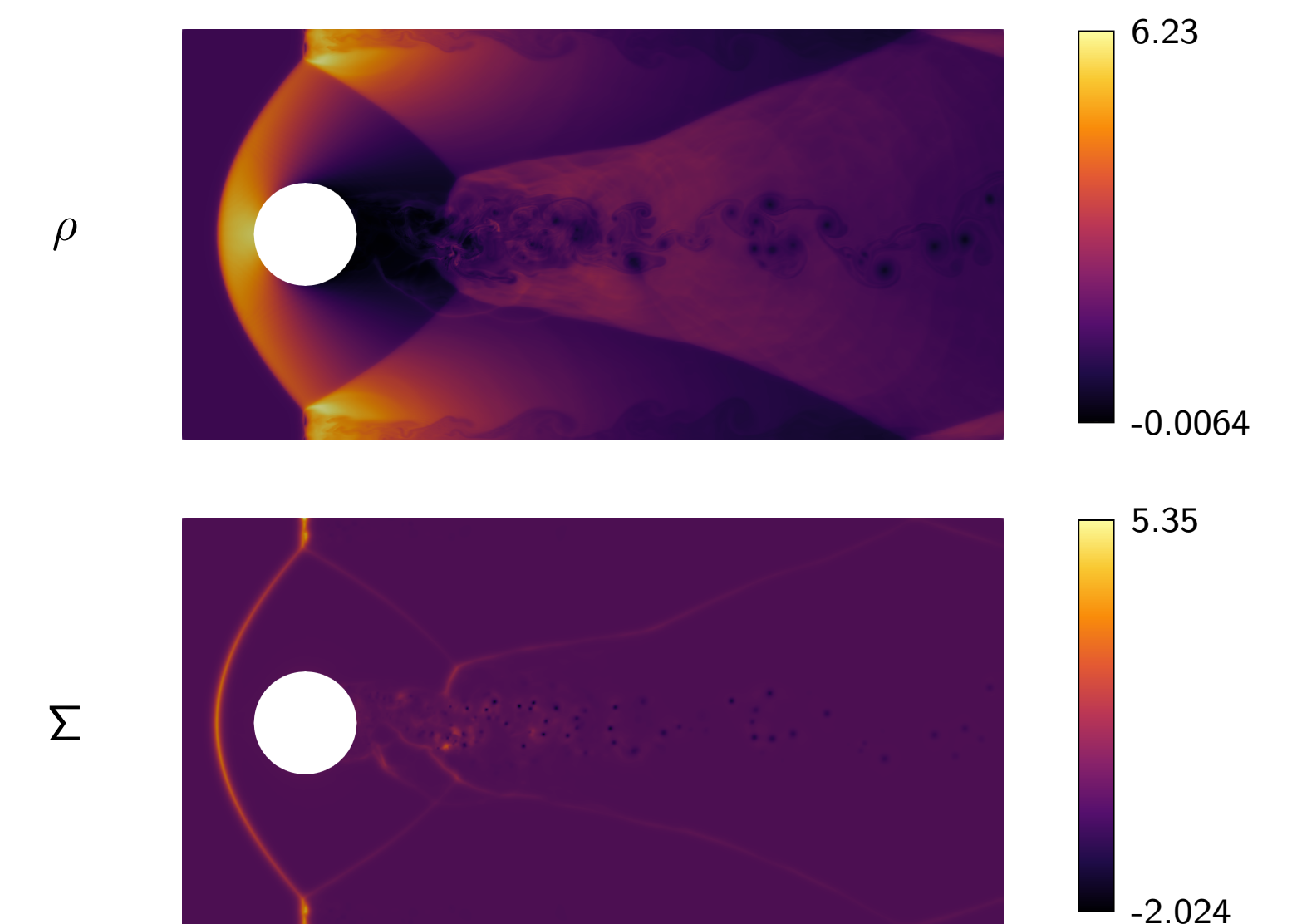


- For fixed DOF, IGR retains more fine-scale structure as p increases without loss of accuracy.



2D Numerical Results: Supersonic Cylinder

- An MFEM [5] implementation extends DG-IGR to higher dimensions.
- Shock structures remain stable without visible mesh-aligned artifacts.
- The supersonic cylinder benchmark generates a detached bow shock with strong localized compression.
- Σ activates near the bow shock while remaining weak in the surrounding flow, preserving fine-scale structure in the freestream and downstream regions.



Two-dimensional supersonic cylinder example with $m = 128 \times 128$ and $p = 2$.

Conclusions

- IGR provides a PDE-level regularization for compressible Euler flow.
- The IGR-DG formulation preserves the conservative DG structure.
- Σ only activates near shocks and strongly compressive regions.
- Compared to other stabilization methods, IGR better preserves fine-scale structure at higher orders for fixed DOF.
- Numerical results demonstrate robust behavior in 1D and 2D compressible flows.

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